

Stock Mate

In February 1996 Gary Kasparov, the reigning world chess champion, faced one of his toughest opponents: Deep Blue, an IBM computer said to be able to sort through 100 million positions a second. The champ won, but it wasn't a KO: the computer took the first game in the six-game match and managed two draws.

In another ten years the machine will probably triumph. Computer chess is getting better over time much faster than people chess is getting better over time. Still, chess is so complex that there is no hope of any human or computer playing "perfect" chess. Humans have done fairly well at it, though. At least, few other creatures can beat them. If you play chess with your dog, you'll beat your dog. The dog's tail will probably knock the pieces off the board at some point, ending the game, but before then it is likely the dog will make some very bad moves.

In simpler games, such as checkers or backgammon, one could design a computer program that would always make a demonstrably "best" move, but this isn't possible in chess because of the enormous number of possible positions reachable in eight or ten future moves. Therefore, a chess program must use a set of heuristic principles to judge a position (control the center, protect the king), and the best chess program is the one with the most useful set of heuristics.

Chess was a good test for a computer, because chess is a game with a fixed number of rules. The rules stay unchanged, and there are no random elements. Because the rules are finite, they can be written into a computer,